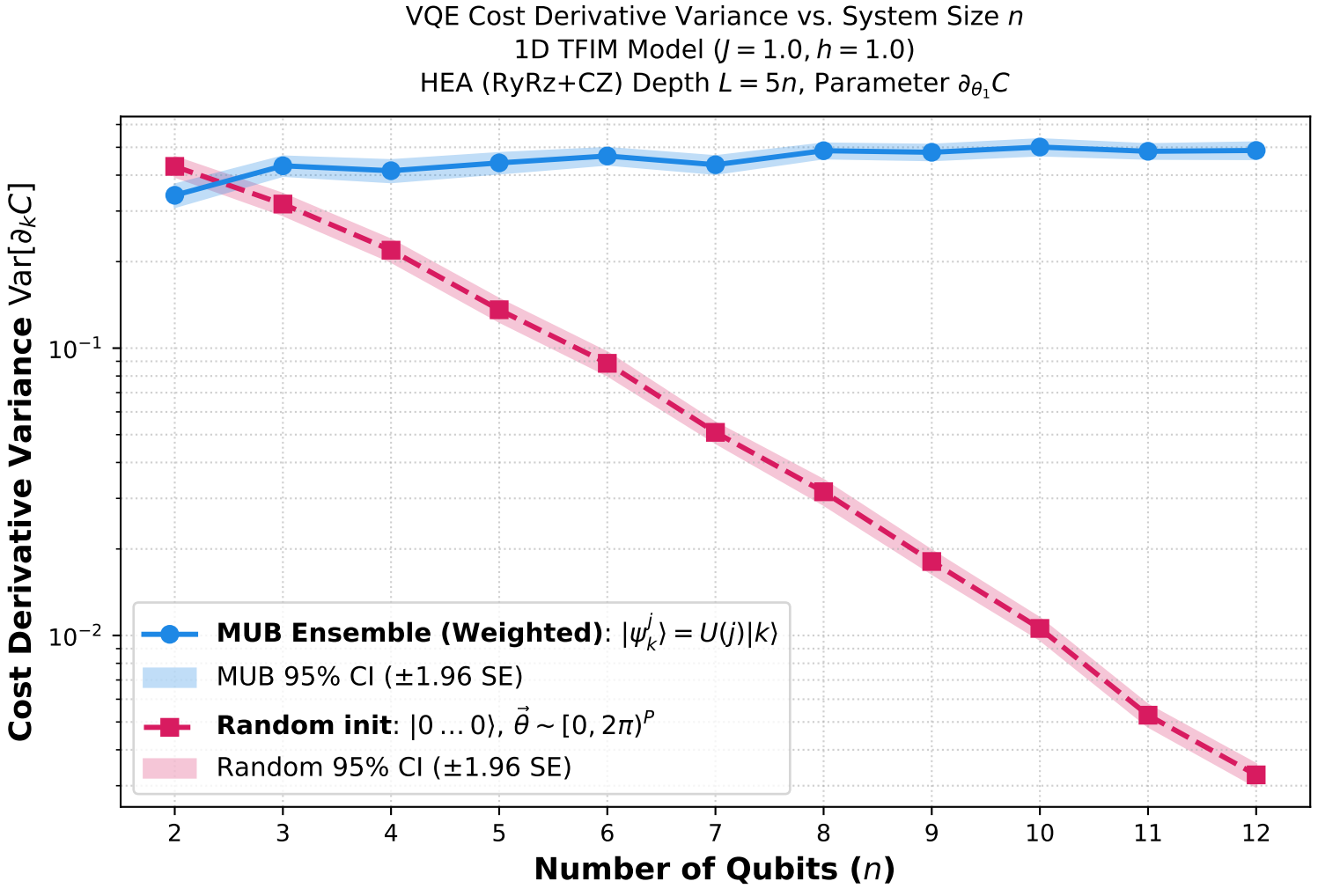
VQE Cost Derivative Variance vs. System Size n
1D TFIM Model ($J = 1.0, h = 1.0$)
HEA (RyRz+CZ) Depth $L = 5n$, Parameter $\partial_{\theta_1} C$
Cost Derivative Variance $\mathrm{Var}[\partial_k C]$
10^{-1}
10^{-2}
MUB Ensemble (Weighted): $|\psi_k^j\rangle = U(j)|k\rangle$
MUB 95% CI (±1.96 SE)
Random init: $|0 \ldots 0\rangle$, $\vec{\theta} \sim [0, 2\pi)^P$
Random 95% CI (±1.96 SE)
2
3
4
5
6
7
8
9
10
11
12
Number of Qubits (n)